
50 STATISTICS QUESTIONS FOR AI / ML INTERVIEWS

Interview-Ready Answers You Can Speak Confidently
No Formula Dumps — Pure Intuition + Real ML Context

HOW EACH ANSWER IS STRUCTURED	
WHAT IT IS	Plain-English definition of the concept
WHY IT MATTERS	Why it is critical in real ML work
SAY IN INTERVIEW	The exact way to impress your interviewer

9 SECTIONS — 50 QUESTIONS — INTERVIEW READY		
1. Descriptive Statistics	2. Probability	3. Probability Distributions
4. Hypothesis Testing	5. Correlation & Regression	6. Estimation & Inference
7. Bayesian Statistics	8. ML-Specific Statistics	9. Advanced Topics

Compiled by a Senior AI/ML Recruiter | 5+ Years of Interview Experience
Speak it. Own it. Clear the interview.

SECTION 1 — DESCRIPTIVE STATISTICS

Q1. What is the difference between Mean, Median, and Mode?

WHAT IT IS Mean is the mathematical average, Median is the exact middle value, and Mode is the most frequently appearing value in the data.

WHY IT MATTERS In real-world data, salary or house prices are skewed — the mean gets pulled by extreme values, so the median tells a more honest story.

SAY IN INTERVIEW "I always check skewness first — if data is skewed, I trust the Median over the Mean because it is not distorted by outliers."

Q2. What is Standard Deviation and why do we care about it?

WHAT IT IS Standard Deviation tells you how spread out data points are around the average — a low value means data is tightly packed, a high value means it is widely scattered.

WHY IT MATTERS Features with very high standard deviation can dominate distance-based ML models like KNN or SVM, which is why we normalize or standardize features before training.

SAY IN INTERVIEW "Before feeding data into any model, I check standard deviation to understand feature spread — it directly guides my preprocessing decisions."

Q3. What is the difference between Population and Sample?

WHAT IT IS Population is every single data point that exists; Sample is the subset you actually have to work with in practice.

WHY IT MATTERS In ML, your training dataset is always a sample — so your model is always estimating population-level patterns, which is why generalization is so hard.

SAY IN INTERVIEW "All of ML is essentially trying to learn population behavior from a sample — that tension between sample and population is what creates overfitting."

Q4. What are Percentiles and the IQR, and how do you use them?

WHAT IT IS Percentiles split your data into 100 equal parts; the IQR (Interquartile Range) is the spread of the middle 50% of your data between the 25th and 75th percentile.

WHY IT MATTERS IQR is the most practical tool for detecting outliers — any point more than 1.5x the IQR above Q3 or below Q1 is flagged as an outlier in the standard box-plot rule.

SAY IN INTERVIEW "I use IQR-based outlier detection during EDA — it is robust to the extremes and does not assume any specific distribution."

Q5. What is Skewness and Kurtosis?

WHAT IT IS Skewness tells you which direction a distribution leans — a long right tail means positive skew, long left tail means negative skew. Kurtosis tells you how heavy or extreme the tails are.

WHY IT MATTERS Highly skewed features hurt linear models and tree-based models can handle them, so knowing this guides your choice of algorithm and transformations like log-transform.

SAY IN INTERVIEW "I always plot distribution and check skewness — if it is strongly skewed, I apply a log or square-root transform before feeding it into linear models."

SECTION 2 — PROBABILITY

Q6. What is the difference between Probability and Likelihood?

WHAT IT IS Probability asks 'given these model parameters, how likely is this data?' — Likelihood flips it and asks 'given this data, how plausible are these parameters?'

WHY IT MATTERS Almost every ML model is trained by maximizing likelihood — logistic regression, neural networks, and Gaussian models all use this principle under the hood.

SAY IN INTERVIEW "Training an ML model is essentially finding the parameters that make the observed training data most likely — that is Maximum Likelihood Estimation."

Q7. What is Conditional Probability and why is it fundamental?

WHAT IT IS Conditional Probability is the probability of event A happening when you already know event B has happened — it narrows your focus to a restricted world where B is true.

WHY IT MATTERS It is the engine behind recommendation systems, spam filters, and medical diagnosis — every time a model updates its prediction based on new evidence, it is doing conditional probability.

SAY IN INTERVIEW "When I explain Naive Bayes to non-technical stakeholders, I say it is just asking — given the words in this email, what is the probability it is spam?"

Q8. What is Bayes' Theorem and why is it so important in ML?

WHAT IT IS Bayes' Theorem is a formula that lets you update your belief about something after seeing new evidence — prior belief combined with new data gives you a refined posterior belief.

WHY IT MATTERS It powers Naive Bayes classifiers, Bayesian optimization for hyperparameter tuning, and is the philosophical backbone of all probabilistic ML models.

SAY IN INTERVIEW "I think of Bayes as the math of learning — you start with a belief, see evidence, and update. That is exactly what ML models do with every batch of training data."

Q9. What is the difference between Independent and Dependent events?

WHAT IT IS Two events are independent if knowing one happened tells you nothing about the other — dependent events are linked, so one changes the probability of the other.

WHY IT MATTERS Naive Bayes assumes features are independent given the class label — this assumption is rarely true in practice but the model still performs surprisingly well.

SAY IN INTERVIEW "The 'naive' in Naive Bayes is the independence assumption — it is wrong almost always, yet the algorithm works well in high dimensions like text classification."

Q10. What is the Law of Total Probability?

WHAT IT IS It says you can find the overall probability of an outcome by breaking the problem into separate scenarios, calculating probability in each, and summing them up weighted by how likely each scenario is.

WHY IT MATTERS This is the mathematical basis for mixture models, hidden Markov models, and the Expectation-Maximization algorithm — all of which sum over hidden or latent variables.

SAY IN INTERVIEW "When a model has hidden states or latent variables, the Law of Total Probability tells you how to marginalize over them — that is the core of the EM algorithm."

SECTION 3 — PROBABILITY DISTRIBUTIONS

Q11. What is the Normal (Gaussian) Distribution and why is it everywhere?

WHAT IT IS A perfectly symmetric bell-shaped distribution defined only by its mean and spread — natural phenomena like height, errors, and measurement noise tend to follow this shape.

WHY IT MATTERS Linear regression assumes residuals are normally distributed; neural network weights are initialized from it; the Central Limit Theorem guarantees that averages of anything become normal.

SAY IN INTERVIEW "The Gaussian is special because it is the attractor — no matter what your data's original shape is, averages of that data converge to Gaussian as sample size grows."

Q12. What is the Binomial Distribution?

WHAT IT IS It models how many successes you get in a fixed number of independent trials where each trial has the same probability of success — like flipping a coin 100 times.

WHY IT MATTERS A/B testing, click-through rate modeling, and binary classification evaluation all rely on binomial assumptions when comparing two groups.

SAY IN INTERVIEW "Every time I run an A/B test, I am modeling conversions as a binomial process — did the user convert or not, across n independent sessions."

Q13. What is the Poisson Distribution?

WHAT IT IS It models the count of rare events happening in a fixed time or space window — like how many users click a button per minute or how many fraud cases occur per day.

WHY IT MATTERS Fraud detection, network packet modeling, and customer arrival rate prediction all use Poisson because these are count-based, low-frequency events.

SAY IN INTERVIEW "If I am predicting number of support tickets per hour, I would model it as Poisson — it is built exactly for counting rare events over a fixed interval."

Q14. What is the Bernoulli Distribution?

WHAT IT IS The simplest distribution — a single coin flip that gives either 1 (success) or 0 (failure) with a fixed probability. Every binary classification prediction is one Bernoulli trial.

WHY IT MATTERS Logistic regression outputs a Bernoulli probability; binary cross-entropy loss is literally derived from the log-likelihood of a Bernoulli distribution.

SAY IN INTERVIEW "When I train a logistic regression model, I am fitting a Bernoulli distribution to each prediction — the loss function is just the negative log of that probability."

Q15. What is the Uniform Distribution?

WHAT IT IS Every outcome in a given range has exactly equal probability — no value is more or less likely than another.

WHY IT MATTERS Random search for hyperparameter tuning samples from uniform distributions; model weight initialization in some networks uses uniform sampling.

SAY IN INTERVIEW "When I do random hyperparameter search, I sample learning rate uniformly on a log scale — no prior assumption about which value is better."

Q16. What is the Exponential Distribution?

WHAT IT IS It models the waiting time between events — like how long until the next user session, the next server crash, or the next customer purchase.

WHY IT MATTERS Survival analysis, churn modeling, and time-to-event prediction all rely on exponential or related distributions to model duration data.

SAY IN INTERVIEW "In churn modeling, I treat the time until a user churns as exponentially distributed — it has the memoryless property, meaning past survival does not affect future risk."

SECTION 4 — HYPOTHESIS TESTING

Q17. What is Hypothesis Testing in simple terms?

WHAT IT IS It is a structured way to use data to decide if an observed effect is real or just happened by random chance — you set up a default assumption and check if data is surprising enough to reject it.

WHY IT MATTERS Every A/B test in a product, every model comparison, and every feature significance check is a hypothesis test — it is the foundation of data-driven decision making.

SAY IN INTERVIEW "I use hypothesis testing to protect against HiPPO decisions — Highest Paid Person's Opinion. The data either supports the change or it does not."

Q18. What is a p-value and what is the most common misconception about it?

WHAT IT IS The p-value is the probability of seeing data this extreme or more extreme, assuming the null hypothesis is actually true — it is NOT the probability that your hypothesis is correct.

WHY IT MATTERS Misinterpreting p-values is one of the most common statistical errors in industry — a small p-value means the result is surprising under the null, not that your model is definitely right.

SAY IN INTERVIEW "The classic mistake is saying $p=0.03$ means there is a 3% chance the null is true — it actually means this data would only appear 3% of the time if the null were true."

Q19. What are Type I and Type II Errors?

WHAT IT IS Type I error is a false alarm — you conclude there is an effect when there really is none. Type II error is a missed detection — you fail to notice a real effect that exists.

WHY IT MATTERS In fraud detection, Type II is more costly (missing real fraud); in medical testing, Type I is more costly (treating healthy patients unnecessarily).

SAY IN INTERVIEW "I always ask the business: which error is more expensive? That determines whether I optimize for precision or recall in my classification model."

Q20. What is Statistical Power?

WHAT IT IS Power is the probability that your test will actually detect a real effect when one truly exists — low power means you could run a study and miss a genuine finding.

WHY IT MATTERS Before launching an A/B test, you should calculate the required sample size to achieve at least 80% power — otherwise you waste resources and get inconclusive results.

SAY IN INTERVIEW "I always do a power analysis before any A/B test — running an underpowered experiment is like looking for your keys under only one streetlight."

Q21. What is the difference between one-tailed and two-tailed tests?

WHAT IT IS A one-tailed test checks if an effect goes only in one specific direction; a two-tailed test checks if there is any difference, in either direction.

WHY IT MATTERS Using a one-tailed test incorrectly inflates your false positive rate — unless you have a strong pre-specified reason to expect a direction, always use two-tailed.

SAY IN INTERVIEW "I default to two-tailed tests in practice — one-tailed tests are only justified when it is theoretically impossible for the effect to go the other way."

Q22. What is the Central Limit Theorem and why does it matter so much in ML?

WHAT IT IS No matter what distribution your individual data points follow, the average of a large enough group of them will follow a Normal distribution — this works for almost any distribution.

WHY IT MATTERS This is why statistical tests, confidence intervals, and many ML assumptions work in practice even when data is not originally normal.

SAY IN INTERVIEW "CLT is the reason we can use normal-based tests on non-normal data when our sample is large enough — it is one of the most powerful free lunches in statistics."

Q23. What is a t-test and when do you use it vs a z-test?

WHAT IT IS Both tests compare means — but the t-test is used when you do not know the population standard deviation and your sample is small; z-test is for large samples or known population variance.

WHY IT MATTERS In practice, you almost never know the true population variance, so t-tests are the default for A/B tests and model comparison experiments.

SAY IN INTERVIEW "I always use a t-test for A/B tests in production — real-world sample sizes are finite and we never know the true population standard deviation."

Q24. What is the Chi-Square Test?

WHAT IT IS It tests whether two categorical variables are related to each other or completely independent — it compares what you actually observed to what you would expect if they were independent.

WHY IT MATTERS Used heavily in feature selection for classification tasks — if a categorical feature is statistically independent of the target label, it is likely useless for the model.

SAY IN INTERVIEW "I use chi-square tests during feature selection for categorical variables — it quickly tells me if a feature carries any signal about the target class."

Q25. What is ANOVA (Analysis of Variance)?

WHAT IT IS ANOVA tests whether the means of three or more groups are different from each other — it does this by comparing how much variation exists between groups versus within groups.

WHY IT MATTERS When comparing performance of multiple ML models or multiple hyperparameter configurations, ANOVA tells you if the differences are statistically significant or just noise.

SAY IN INTERVIEW "When I compare more than two models, I use ANOVA instead of multiple t-tests — running multiple t-tests inflates the false positive rate."

SECTION 5 — CORRELATION & REGRESSION

Q26. What is the difference between Correlation and Causation?

WHAT IT IS Correlation means two variables move together statistically; causation means one variable actually causes the other to change — correlation never proves causation.

WHY IT MATTERS ML models detect correlation, not causation — a model trained on correlated data can fail badly when the correlation breaks in production due to a real-world change.

SAY IN INTERVIEW "Ice cream sales correlate with drowning rates — both rise in summer. A model would learn this correlation, but deploying it to reduce drowning by banning ice cream would be wrong."

Q27. What is Pearson vs Spearman Correlation?

WHAT IT IS Pearson measures how linearly two variables move together; Spearman measures whether they move together monotonically (same direction) even if the relationship is curved.

WHY IT MATTERS Pearson is sensitive to outliers and assumes linearity; Spearman is robust and works for any monotonic relationship — crucial when data is skewed or has extreme values.

SAY IN INTERVIEW "I start with Spearman during EDA because it is robust — if Spearman and Pearson give very different results, that tells me there are outliers or non-linearity to investigate."

Q28. What is Multicollinearity and how do you deal with it?

WHAT IT IS Multicollinearity happens when two or more features in your dataset are so highly correlated that the model cannot distinguish their individual contributions to the target.

WHY IT MATTERS It makes model coefficients unstable and uninterpretable — a tiny change in data flips the coefficient sign, destroying any business insight from the model.

SAY IN INTERVIEW "I check VIF scores during feature engineering — if two features have VIF above 10, I either drop one, combine them, or use Ridge regression to let regularization handle it."

Q29. What are the key assumptions of Linear Regression?

WHAT IT IS Linear regression assumes the relationship is linear, errors are independent of each other, spread of errors is consistent (not funnel-shaped), and errors roughly follow a bell curve.

WHY IT MATTERS Violating these assumptions makes your predictions unreliable and your confidence intervals meaningless — always validate residual plots before trusting a regression model.

SAY IN INTERVIEW "I always plot residuals vs fitted values — a funnel shape tells me heteroscedasticity, a curve tells me non-linearity, and both signal I need to transform features."

Q30. What is R-squared and when is it misleading?

WHAT IT IS R-squared tells you what percentage of variation in the target your model explains — but it always increases when you add more features, even useless ones.

WHY IT MATTERS Adjusted R-squared penalizes for extra features — always use it when comparing models with different numbers of variables, or you will fool yourself into thinking more features = better model.

SAY IN INTERVIEW "R-squared alone is a vanity metric — I always pair it with Adjusted R-squared and error metrics like RMSE to get an honest picture of model quality."

SECTION 6 — ESTIMATION & INFERENCE

Q31. What is Maximum Likelihood Estimation (MLE) intuitively?

WHAT IT IS MLE is the process of finding model parameters that make the data you observed the most probable — it is like adjusting model dials until the observed data becomes the most expected outcome.

WHY IT MATTERS Logistic regression, neural networks, and Gaussian models are all trained using MLE under the hood — understanding this connects loss functions to probabilistic thinking.

SAY IN INTERVIEW "Cross-entropy loss in neural networks is actually negative log-likelihood — minimizing loss is the same as maximizing likelihood. Loss functions are not arbitrary; they are probabilistic."

Q32. What is the difference between MLE and MAP Estimation?

WHAT IT IS MLE uses only the data to estimate parameters; MAP adds a prior belief about what the parameters should look like and combines it with the data.

WHY IT MATTERS Adding a Gaussian prior in MAP is mathematically identical to L2 (Ridge) regularization — this connects Bayesian thinking directly to the regularization tricks used in ML.

SAY IN INTERVIEW "When I use Ridge regression, I am actually doing MAP estimation with a Gaussian prior — regularization is Bayesian thinking in disguise."

Q33. What is a Confidence Interval and what does it really mean?

WHAT IT IS A 95% confidence interval does NOT mean there is a 95% chance the true value is inside — it means if you repeated the study 100 times, about 95 of those intervals would contain the true value.

WHY IT MATTERS Misunderstanding confidence intervals leads to overconfident business decisions — a wide interval means your estimate is very uncertain, not just imprecise.

SAY IN INTERVIEW "I always report confidence intervals alongside point estimates to stakeholders — a narrow CI means the model is reliable; a wide CI means we need more data before acting."

Q34. What is the Bias-Variance Tradeoff?

WHAT IT IS Bias is when your model is too simple and consistently wrong in the same direction; Variance is when your model is too complex and gives wildly different answers on different data.

WHY IT MATTERS Every ML modeling decision — regularization strength, tree depth, number of layers — is essentially tuning the bias-variance balance for your specific problem.

SAY IN INTERVIEW "A decision tree that is too deep memorizes the training data — high variance. A linear model on non-linear data is always wrong in the same way — high bias. I tune regularization to balance both."

Q35. What is Overfitting and how do you detect and fix it?

WHAT IT IS Overfitting is when your model learns the noise and quirks of training data so well that it performs great on training data but terribly on new, unseen data.

WHY IT MATTERS A model that overfits has zero real-world value — it is essentially memorizing the training set, not learning generalizable patterns.

SAY IN INTERVIEW "I always plot train vs validation loss curves — if training loss keeps dropping but validation loss starts rising, that is the exact moment overfitting begins and I stop training there."

SECTION 7 — BAYESIAN STATISTICS

Q36. What is the difference between Frequentist and Bayesian thinking?

WHAT IT IS Frequentists treat data as random and parameters as fixed unknowns; Bayesians treat parameters as uncertain and update their probability distributions as new data arrives.

WHY IT MATTERS Bayesian methods naturally handle small datasets, incorporate domain knowledge as priors, and give uncertainty estimates — all critical needs in production ML systems.

SAY IN INTERVIEW "Bayesian approaches are powerful for small-data problems — instead of needing thousands of examples, you inject domain expertise as a prior and let data refine the estimate."

Q37. What is a Prior, Likelihood, and Posterior in plain English?

WHAT IT IS Prior is what you believe before seeing any data; Likelihood is how well the data fits a given hypothesis; Posterior is your updated belief after combining both.

WHY IT MATTERS This cycle of update is how humans learn and how Bayesian ML systems improve over time — every new batch of data shifts the posterior, which becomes the prior for the next update.

SAY IN INTERVIEW "I describe Bayesian updating as a doctor's reasoning: your prior is medical knowledge, the test result is the likelihood, and your diagnosis is the posterior — updated belief."

Q38. What is MCMC (Markov Chain Monte Carlo) conceptually?

WHAT IT IS MCMC is a smart random-sampling technique used when a probability distribution is too complex to compute directly — it explores the distribution by taking guided random steps.

WHY IT MATTERS Bayesian deep learning, probabilistic programming, and sampling from complex posteriors all rely on MCMC when exact computation is mathematically intractable.

SAY IN INTERVIEW "Think of MCMC as a drunk man exploring a hilly landscape — over time, he spends more time in valleys (high-probability regions). His path traces out the distribution."

SECTION 8 — ML-SPECIFIC STATISTICS

Q39. What is Cross-Validation and why is a single train-test split not enough?

WHAT IT IS Cross-validation trains and tests the model multiple times on different portions of the data, then averages the results to get a reliable performance estimate.

WHY IT MATTERS A single split gets lucky or unlucky depending on which data points end up in the test set — cross-validation removes that randomness and gives a stable, trustworthy score.

SAY IN INTERVIEW "I always use k-fold cross-validation for model selection — a single split is like judging a restaurant based on one visit. Multiple visits give a fairer picture."

Q40. What is the difference between L1 and L2 Regularization intuitively?

WHAT IT IS L1 (Lasso) pushes some feature weights all the way to zero — automatically selecting the most important features; L2 (Ridge) shrinks all weights towards zero but rarely eliminates any.

WHY IT MATTERS Use L1 when you suspect most features are irrelevant and want automatic feature selection; use L2 when all features likely matter but are noisy or multicollinear.

SAY IN INTERVIEW "L1 is like a strict editor who cuts whole sentences; L2 is like one who just tones down every word. I use L1 when I want a sparse, interpretable model."

Q41. What is the ROC Curve and AUC, and when is AUC misleading?

WHAT IT IS ROC curve plots the model's true positive rate against its false positive rate at every possible decision threshold; AUC summarizes the whole curve as a single score from 0 to 1.

WHY IT MATTERS AUC is misleading on highly imbalanced datasets — a model that predicts the majority class 100% of the time can still get high AUC; Precision-Recall AUC is better in that case.

SAY IN INTERVIEW "On fraud detection with 0.1% fraud rate, I use PR-AUC instead of ROC-AUC — ROC looks great even for terrible models when classes are imbalanced."

Q42. What is Precision, Recall, and F1, and when do you prioritize each?

WHAT IT IS Precision is about not raising false alarms; Recall is about not missing real cases; F1 is the balanced score that is only high when both are high simultaneously.

WHY IT MATTERS In cancer screening, missing a real case (low recall) is worse than a false alarm — so maximize recall. In spam filtering, flagging real emails as spam (low precision) is worse — so maximize precision.

SAY IN INTERVIEW "I always ask the business: what costs more — a false positive or a false negative? That answer directly determines whether I optimize for precision, recall, or F1."

Q43. What is the difference between Parametric and Non-Parametric tests?

WHAT IT IS Parametric tests make strong assumptions about the distribution of your data (usually Normal); non-parametric tests make no distribution assumptions and work on ranks instead.

WHY IT MATTERS In real-world ML, data is messy and rarely normally distributed — knowing when to use non-parametric alternatives prevents you from drawing wrong statistical conclusions.

SAY IN INTERVIEW "If I cannot verify normality or have small sample sizes, I switch to Mann-Whitney U instead of t-test — it is more honest even if slightly less powerful."

Q44. What is Bootstrapping and how does it power Random Forests?

WHAT IT IS Bootstrapping means repeatedly resampling your dataset with replacement to create many slightly different versions of your data, then analyzing variation across those samples.

WHY IT MATTERS Random Forests train each tree on a bootstrapped sample — this forces diversity among trees, reduces overfitting, and is why the ensemble performs so much better than any single tree.

SAY IN INTERVIEW "The 'bag' in bagging is bootstrap aggregation — each tree sees a different bootstrapped version of the data, so the trees disagree usefully rather than all making the same mistakes."

Q45. What is Effect Size and why is statistical significance not enough?

WHAT IT IS Effect size measures how large or meaningful a difference actually is in practical terms — not just whether it is statistically detectable given your sample size.

WHY IT MATTERS With millions of users, even a 0.001% difference in conversion rate becomes statistically significant, but it is completely meaningless for the business.

SAY IN INTERVIEW "Statistical significance tells you the effect is real; effect size tells you if it matters. I always report both — significant but tiny effects do not justify product changes."

SECTION 9 — ADVANCED TOPICS

Q46. What is PCA (Principal Component Analysis) in plain English?

WHAT IT IS PCA finds the directions in your high-dimensional data along which the data varies the most, then compresses the data along those directions, discarding the low-variance noise.

WHY IT MATTERS It removes multicollinearity, reduces training time, prevents overfitting in high dimensions, and helps visualize clusters in 2D or 3D — all critical in real ML workflows.

SAY IN INTERVIEW "I use PCA when I have 500 features and many are correlated — I reduce to 50 components that capture 95% of variance. The model trains faster and often generalizes better."

Q47. What is the Curse of Dimensionality?

WHAT IT IS As you add more features to your data, the data becomes increasingly sparse in that high-dimensional space — points that seem close in 3D become far apart in 300 dimensions.

WHY IT MATTERS Distance-based algorithms like KNN and clustering degrade badly in high dimensions — all points start looking equally far from each other, so 'nearest neighbor' becomes meaningless.

SAY IN INTERVIEW "The Curse of Dimensionality is why KNN fails silently at scale — in 1000 dimensions, every point is the 'farthest neighbor' of every other. I always reduce dimensions first."

Q48. What is Entropy and Information Gain in Decision Trees?

WHAT IT IS Entropy measures how mixed or uncertain a group of labels is — a node with 50/50 split has maximum entropy (most uncertain); a node with all one class has zero entropy (perfectly certain).

WHY IT MATTERS Decision trees split data on the feature that reduces entropy the most — this is Information Gain, and it is the algorithm's mechanism for learning which features are most predictive.

SAY IN INTERVIEW "A decision tree is essentially asking at each step: which feature, if I split on it, makes my groups the most pure? Entropy is the measure of impurity it is trying to minimize."

Q49. What is the Law of Large Numbers and how does it apply to ML?

WHAT IT IS As you collect more data, your sample statistics (like the mean) will converge toward the true population value — the more data you have, the more trustworthy your estimates become.

WHY IT MATTERS This is the theoretical guarantee that more training data improves ML models — it is why data collection is often more valuable than model architecture tuning.

SAY IN INTERVIEW "The Law of Large Numbers is why 'more data beats better algorithms' is a widely repeated truth in ML — eventually, the data speaks loudly enough to overcome model imperfection."

Q50. What is A/B Testing and what are the most common mistakes people make?

WHAT IT IS A/B Testing is a controlled experiment that randomly splits users into two groups, exposes each to a different version, and uses statistical tests to determine which version performs better.

WHY IT MATTERS The most common mistakes are stopping early when you see a good result (peeking bias), running tests with too few users (underpowered), and testing too many things at once (multiple comparison problem).

SAY IN INTERVIEW "I always pre-calculate required sample size before launching any A/B test and commit to running it fully — peeking at results early and stopping is the most dangerous mistake in experimentation."

5 GOLDEN RULES FOR STATS IN ML INTERVIEWS

#	RULE	WHY IT WINS THE INTERVIEW
1	Always connect stats to ML context	Shows you are a practitioner, not just a textbook reader
2	Lead with intuition, then go deeper	Interviewers test understanding, not formula memorization
3	Know the failure modes of every method	Shows you have used it in real life and hit real problems
4	Mention trade-offs without being asked	Signals senior-level maturity and intellectual honesty
5	Use concrete examples from past projects	Makes your answer memorable and proves real experience

YOU ARE READY. WALK IN WITH CONFIDENCE AND SPEAK WITH CLARITY.
